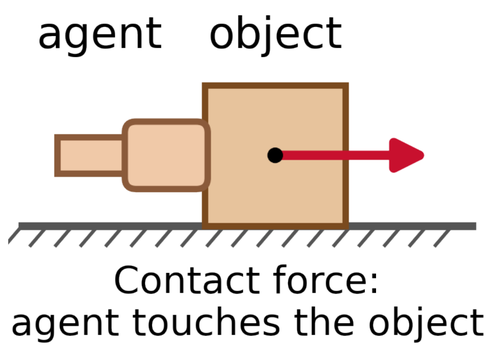
Non-uniform circular motion

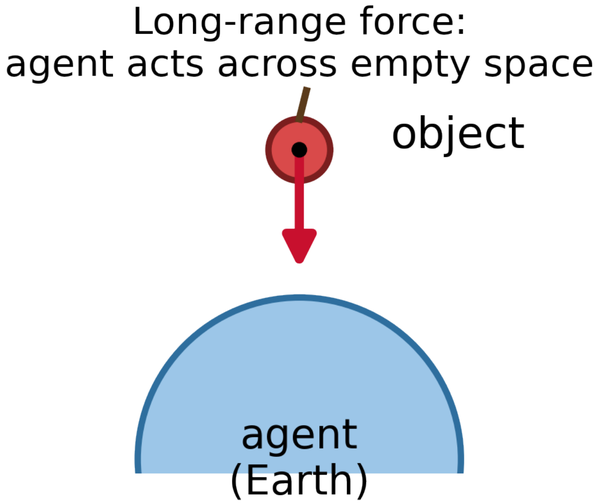
- Starting from rest, a DVD steadily accelerates to $500\,\mathrm{rpm}$ in $1.0\,\mathrm{s}$, rotates at this angular speed for $3.0\,\mathrm{s}$, then steadily decelerates to a halt in $2.0\,\mathrm{s}$. How many revolutions does it make?

Force and Motion

What is a force?

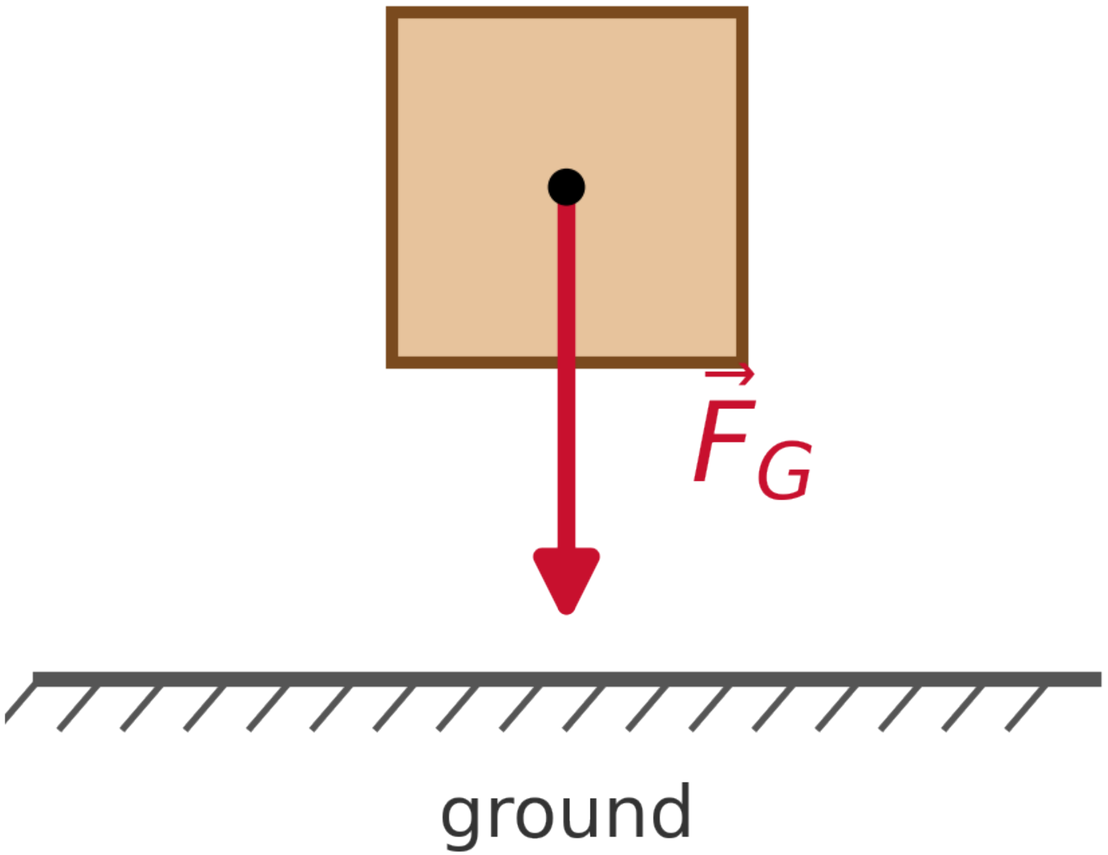
- A force is a push or pull
- Every force acts on an object — something is being pushed or pulled
- Every force has an agent — something is doing the push or pull
- Forces come in two kinds:
  - Contact force — agent touches the object (a hand, a rope, a surface)
  - Long-range force — agent acts from a distance (gravity, magnets)
- A force is a vector — draw force from object pointing in direction of push or pull





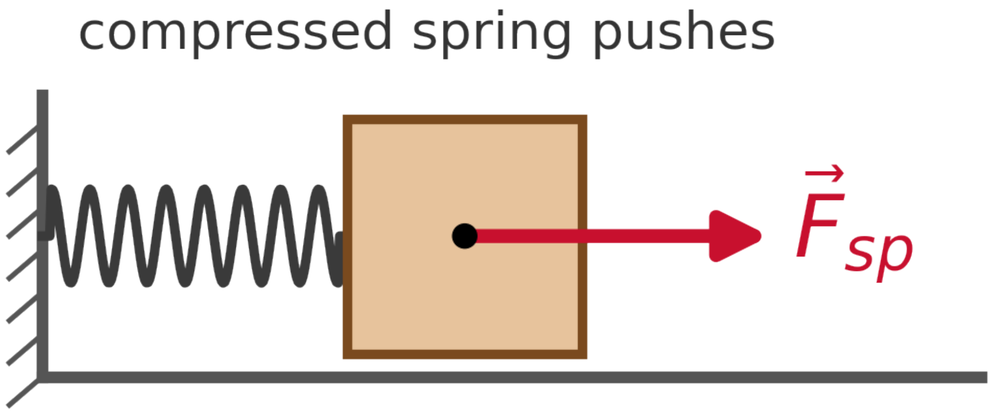
A short catalog of forces

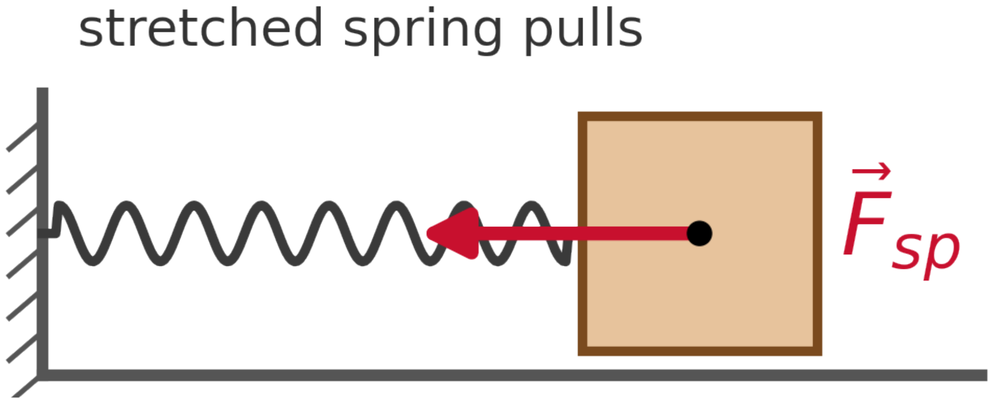
- Gravity $\vec{F}_G$
  - Agent: entire planet (e.g. Earth)
  - Acts on: any object on or near the surface
  - Long-range force
  - Direction: straight down (toward center of planet)



A short catalog of forces

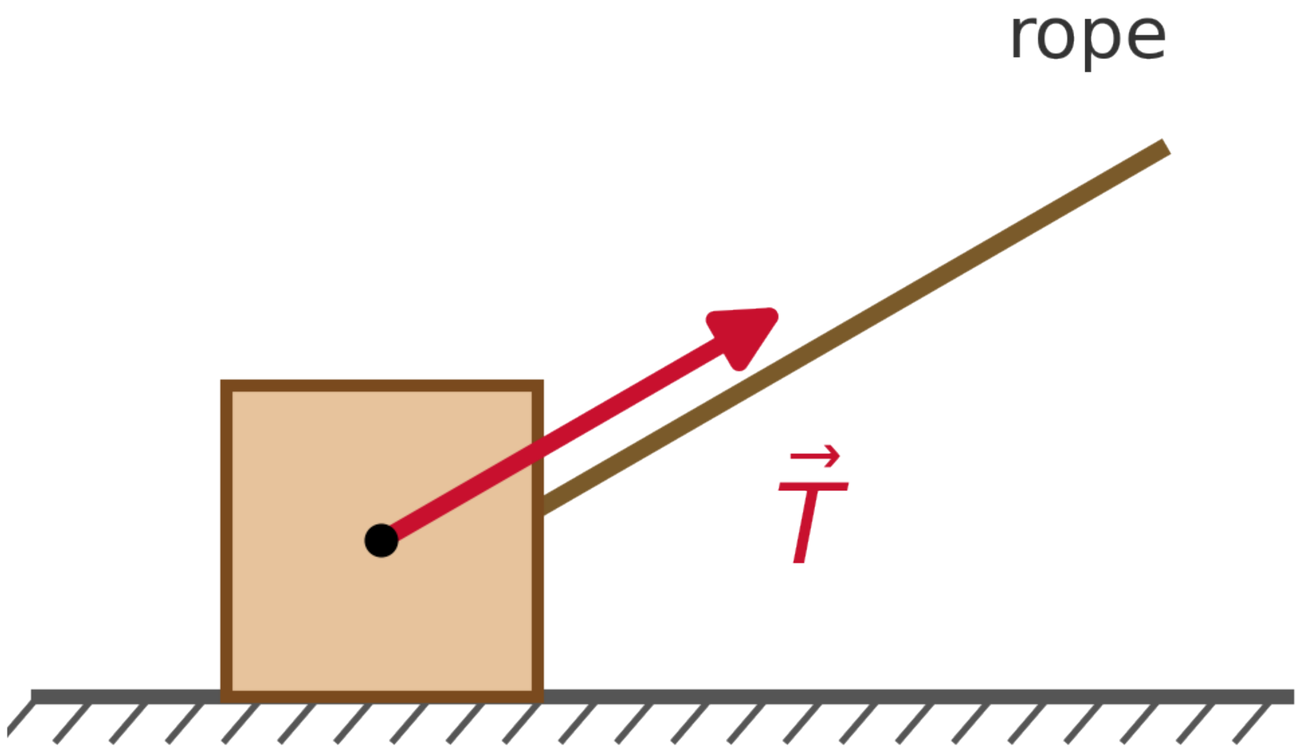
- Spring force $\vec{F}_{\text{sp}}$
  - Agent: spring
  - Acts on: object attached to spring
  - Contact force
  - Direction: along the spring; pushes when compressed, pulls when stretched
  - Always tries to return to its natural length





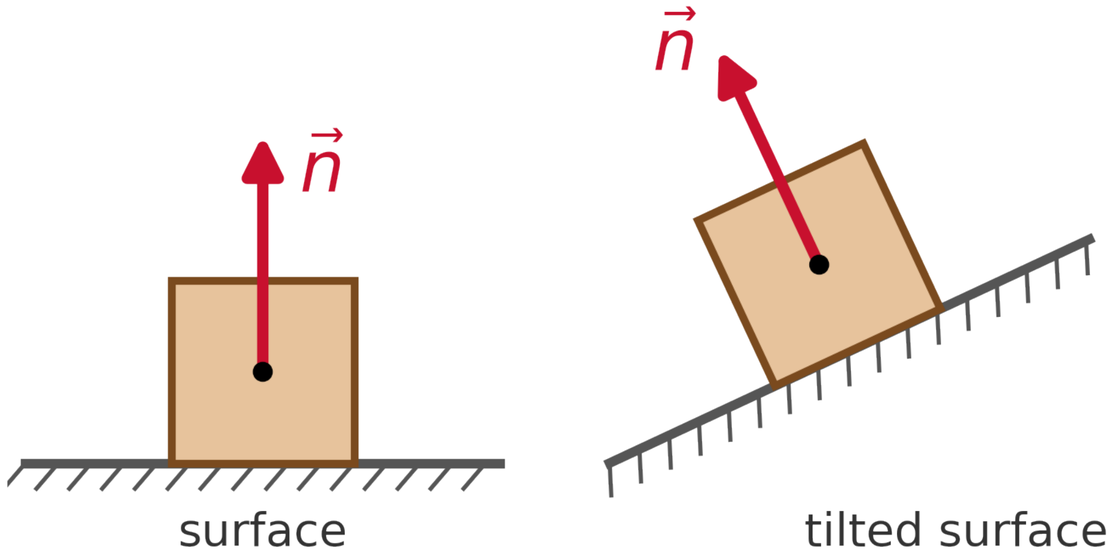
A short catalog of forces

- Tension $\vec{T}$
  - Agent: string, rope, cable, wire, etc.
  - Acts on: object attached to rope
  - Contact force
  - Direction: along the rope
  - Ropes can only pull, never push



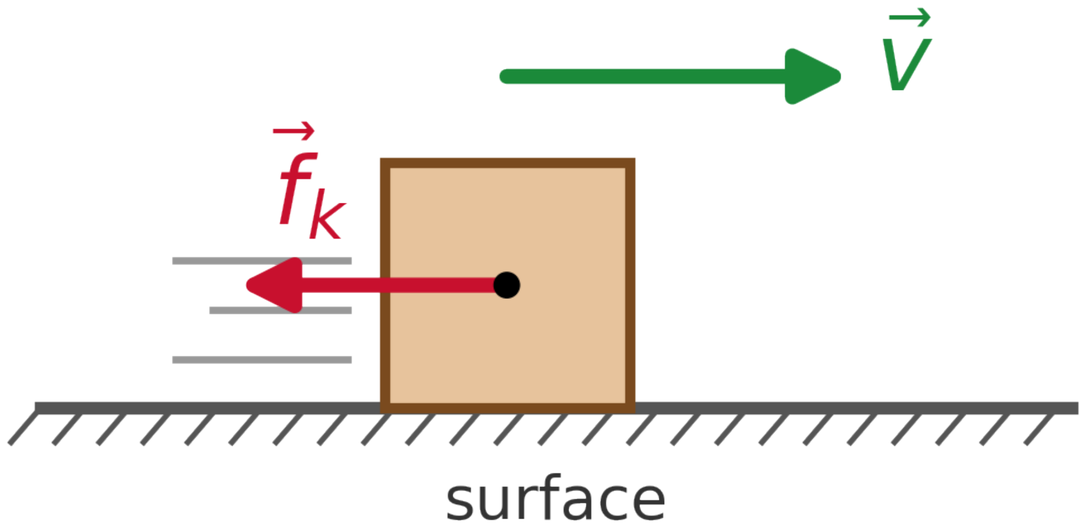
A short catalog of forces

- Normal force $\vec{n}$
  - Agent: a surface
  - Acts on: the object pressing against the surface
  - Contact force
  - Direction: perpendicular (“normal”) to the surface, pushing away from it
  - Surface pushes back just hard enough to keep object from sinking in surface



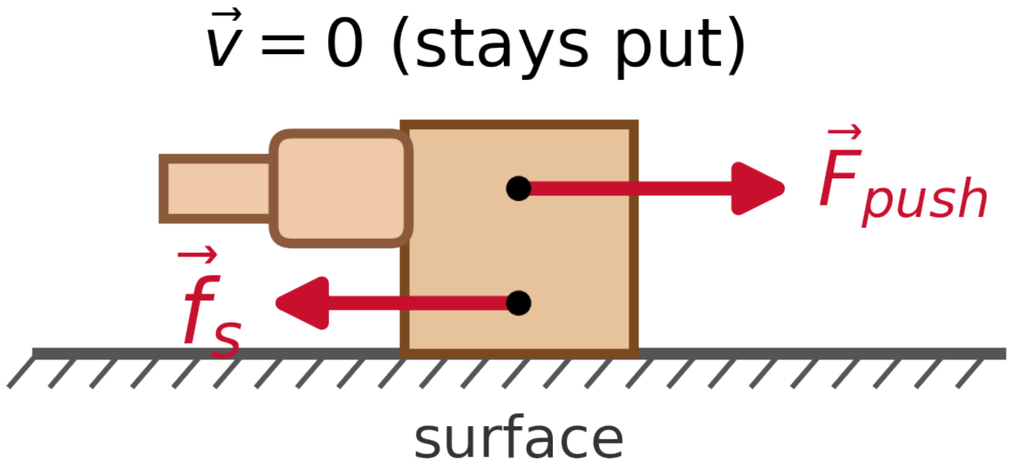
A short catalog of forces

- Kinetic friction $\vec{f}_{\mathrm{k}}$
  - Agent: surface
  - Acts on: object sliding across the surface
  - Contact force
  - Direction: parallel to the surface, opposite the velocity (“opposes the motion”)
  - Always slows sliding down



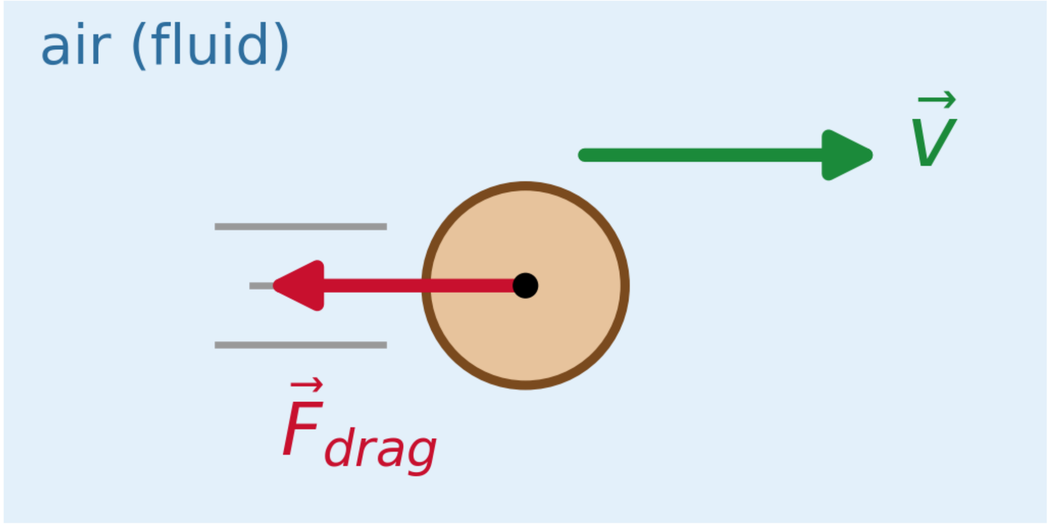
A short catalog of forces

- Static friction $\vec{f}_{\mathrm{s}}$
  - Agent: surface
  - Acts on: an object at rest on the surface that is being pushed or pulled
  - Contact force
  - Direction: parallel to the surface, opposite the way the object would slide if there were no friction
  - Keeps object stuck
  - Magnitude adjusts up to a maximum



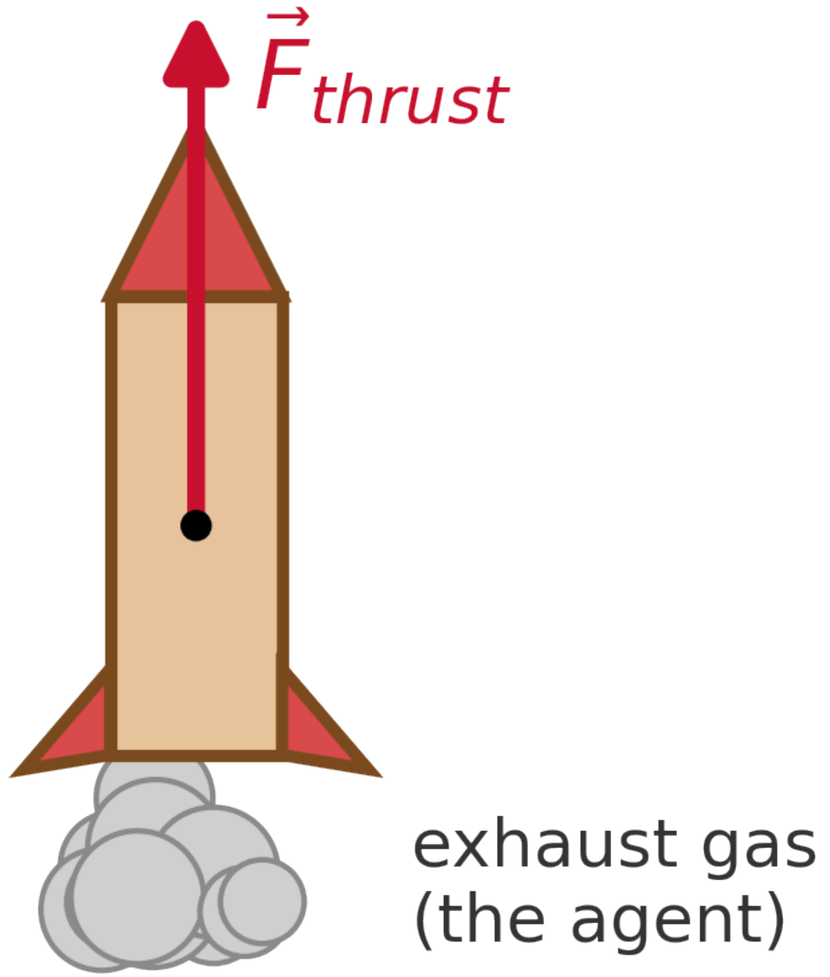
A short catalog of forces

- Drag $\overrightarrow{F}_{\text{drag}}$
  - Agent: a fluid (air, water)
  - Acts on: object moving through the fluid
  - Contact force
  - Direction: opposite the velocity
  - Example: air resistance



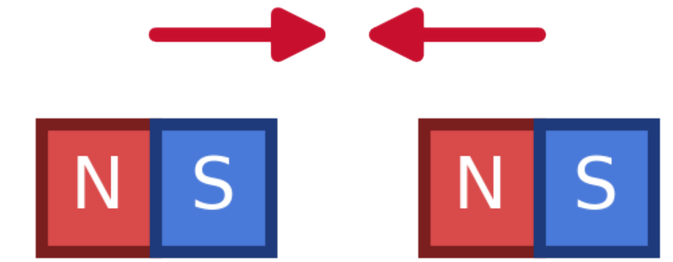
A short catalog of forces

- Thrust $\overrightarrow{F}_{\text{thrust}}$
  - Agent: exhaust gas being expelled
  - Acts on: the engine (rocket, jet)
  - Contact force
  - Direction: opposite the direction the gas is expelled

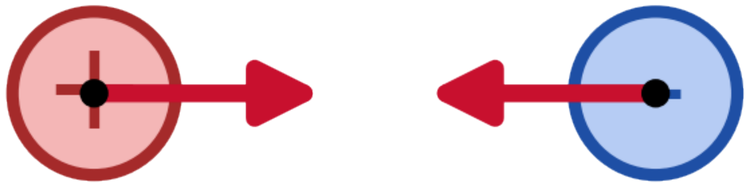


A short catalog of forces

- Electric and magnetic forces
  - Agent: charges objects or magnets
  - Acts on: other charged objects or magnets
  - Long-range
  - Topic of PHYS 2510; not covered here



magnetic force



electric force

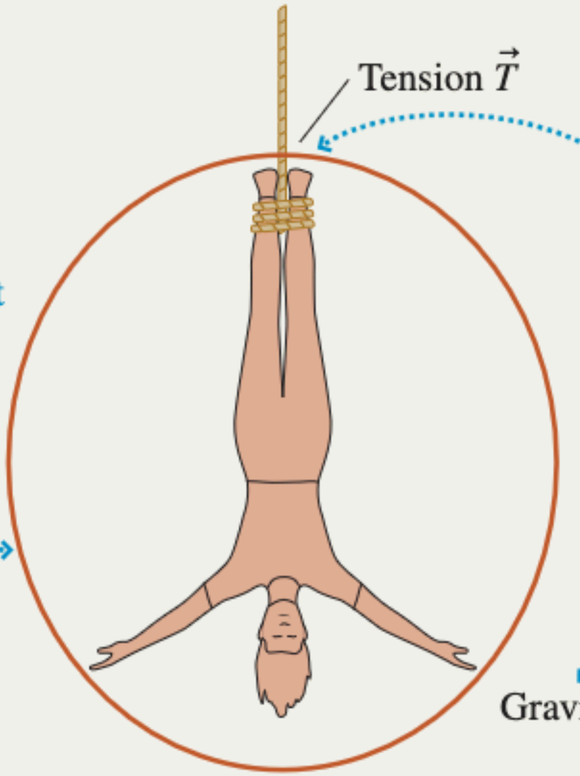
Identifying forces

- Steps
  - Identify object of interest (object you want to study)
  - Draw picture of the situation
  - Draw a closed curve around the object (only object of interest is inside the curve, everything else outside
  - Locate everything point on boundary of curve where other objects touch the object of interest (these are the points where contact forces are exerted on object)
  - Name and label each contact force acting on the object (at least one force at each point of contact)
  - name and label each long-range force acting on object (for now, this is only gravity)

EXAMPLE 5.1 ■ Forces on a bungee jumper

A bungee jumper has leapt off a bridge and is nearing the bottom of her fall. What forces are being exerted on the jumper?

VISUALIZE **FIGURE 5.11** Forces on a bungee jumper.



❶ Identify the object of interest. Here the object is the bungee jumper.

❷ Draw a picture of the situation.

❸ Draw a closed curve around the object.

❹ Locate the points where other objects touch the object of interest. Here the only point of contact is where the cord attaches to her ankles.

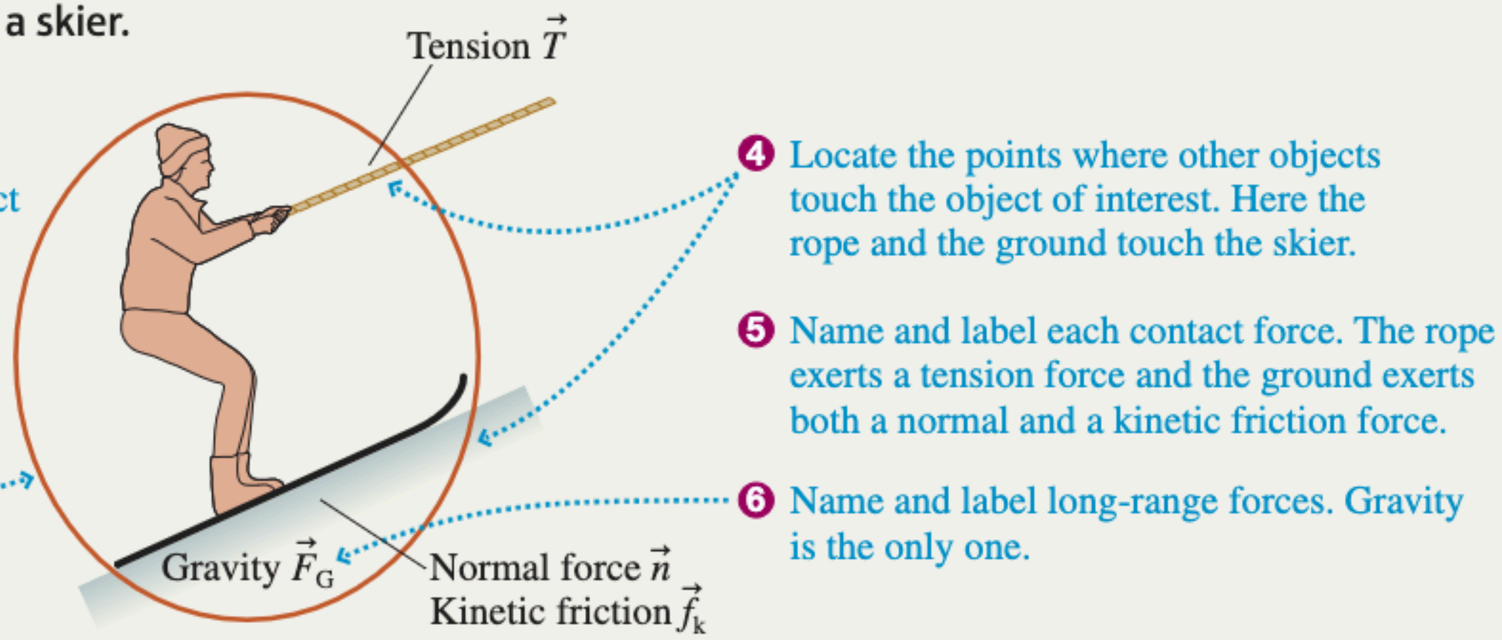
❺ Name and label each contact force. The force exerted by the cord is a tension force.

❻ Name and label long-range forces. Gravity is the only one.

EXAMPLE 5.2 ■ Forces on a skier

A skier is being towed up a snow-covered hill by a tow rope. What forces are being exerted on the skier?

VISUALIZE **FIGURE 5.12** Forces on a skier.



What do forces do?

- Forces make objects move
- Constant forces make objects move with constant acceleration
- The acceleration is directly proportional to the force and inversely proportional to the object's mass

$$\vec{a} = \frac{\vec{F}}{m}$$

- SI unit for force "newton" N — force required to accelerate a $1\,\mathrm{kg}$ mass to $1\,\mathrm{m/s^2}$

$$1\,\mathrm{N} = 1\,\frac{\mathrm{kg\,m}}{\mathrm{s}^2}$$

Combining forces

- Object likely to be subjected to several forces $\vec{F}_1, \vec{F}_2, \vec{F}_3, \ldots$ all with different magnitudes and and directions

- The total force or “net force” is:

$$\vec{F}_{\text{net}} = \sum_{i=1}^{N} \vec{F}_i = \vec{F}_i + \vec{F}_2 + \ldots + \vec{F}_N$$

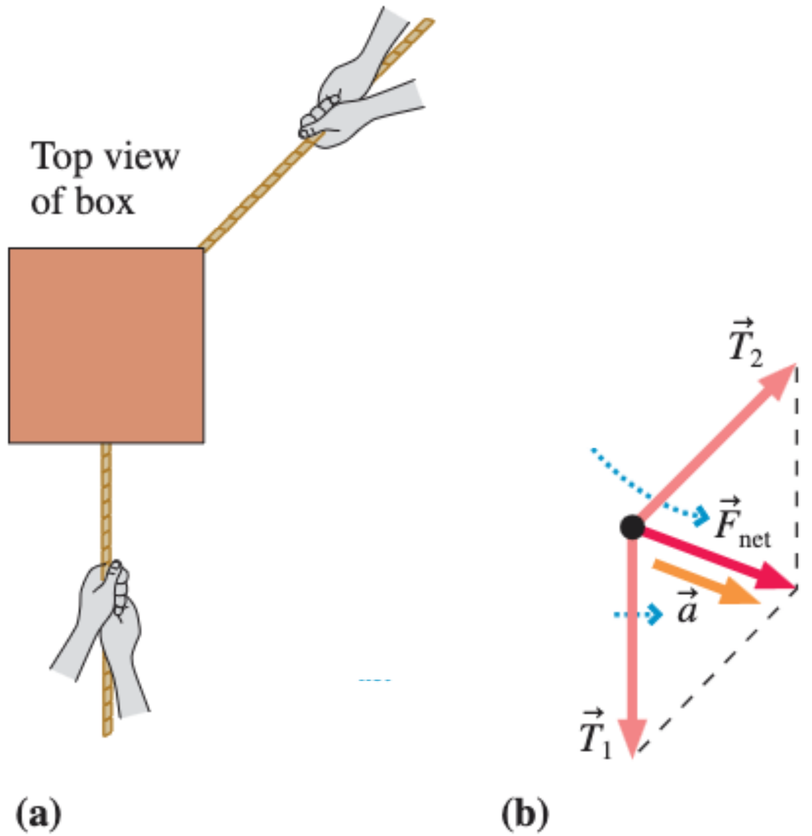
- “superposition of forces”

Newton's second law

An object of mass m subjected to forces $\vec{F}_1, \vec{F}_2, \vec{F}_3, \ldots$ will undergo an acceleration $\vec{a}$ given by $\vec{a} = \dfrac{\vec{F}_{\text{net}}}{m}$ where $\vec{F}_{\text{net}}$ is the vector sum of all forces attacking on the object.

Newton’s second law

- The critical ideas: object accelerates in the direction of the net force vector $\overrightarrow{F}_{\text{net}}$
- $\vec{a}$ and $\overrightarrow{F}_{\text{net}}$ point in the same direction
- Newton’s second law also written as $\overrightarrow{F}_{\text{net}} = m\vec{a}$
- Object responds only to forces acting now not in past (instantaneous)



Mass

- Mass is
  - the amount of matter an object contains
  - how a object responds to a force
- The more matter an object has the more it resists accelerating in response to a force
- **inertia** — The property of an object that makes it resist changes in its velocity.
- **Inertial mass** — A number that quantifies how much inertia an object has. Equation to $m = |\vec{F}_{\text{net}}| / |\vec{a}|$ (because they both point in the same direction)

Forces as interactions

- In general, when an agent exerts a force on an object, the object exerts a force on the agent (**Newton's third law** — more next chapter)

Newton's first law

- "Law of inertia"

  An object at rest will remain at rest; an object that is moving will continue to move in a straight line with constant velocity, if and only if the net force acting on it is zero

- **Net force is zero** — the object may still have forces exerted on it

- "if and only if"

  - if an object is moving with constant velocity in a straight line then $\vec{F}_{\text{net}} = 0$

  - If $\vec{F}_{\text{net}} = 0$ then $\vec{v} = \text{constant}$ (i.e., direction and magnitude!)

Terms

- Mechanical equilibrium — and object at rest or moving in a straight line with constant velocity, i.e., an object with zero net force.

Three critical criteria for identifying forces

- A force has an agent.
- Forces exist at the point of contact between agent and object (expect for a few special cases of long-range forces)
- Forces exist due to interactions happening now, not the past

Think about it

- You’re in a car as a passenger going at normal highway speeds, and suddenly the car stops hard. What happens to you? What do you experience?

Inertial reference frames

- A force is a push or pull **caused by an identifiable agent**
- A acceleration without an apparent force comes from using an inappropriate reference frame.
- An inertial reference frame is one in which Newton's first law is valid.
  - If $\vec{a} = 0$ only when $\overrightarrow{F}_{\text{net}} = 0$, then the reference frame is an inertial reference frame.
  - **Accelerating reference frames are not inertial reference frames.**
- Car example: in this case your acceleration is measured in the reference frame attached to the car which is not the same as your acceleration measured in a reference frame attached to the ground.

Free-body diagrams

- A part of the pictorial representation of a problem
- Steps
  - Identify all forces acting on the object
  - Draw a coordinate system (use axes that match the direction of motion)
  - Represent the object as a dot at the origin of the coordinate axes
  - Draw vectors representing each of the identified forces.
  - Draw and label the net force vector $\overrightarrow{F}_{\text{net}}$ (draw this vector beside the diagram, not on the particles or write $\overrightarrow{F}_{\text{net}} = 0$. Make sure the acceleration vector $\vec{a}$ of the motion diagram points in the same direction as $\overrightarrow{F}_{\text{net}}$
- Next chapter will write equations of motion directly from the free-body diagram

An elevator, suspended by a cable, speeds up as it moves upward from the ground floor. Identify the forces and draw a free-body diagram of the elevator.

MODEL Model the elevator as a particle.

VISUALIZE

FIGURE 5.21 Free-body diagram of an elevator accelerating upward.

